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Modeling and Interpretation of Scattering Mechanisms in Polarimetric Synthetic Aperture Radar

Advances and perspectives

Polarimetric target decomposition is a powerful technique to interpret scattering mechanisms in polarimetric synthetic aperture radar (PolSAR) data. Eigenvalue-eigenvector-based and model-based methods are two main categories within the incoherent decomposition techniques. Eigenvalue-eigenvector-based decomposition becomes relatively mature since it has a clearer mathematical background and has only one decomposition solution. In contrast, model-based decompositions can obtain different decomposition solutions in terms of various scattering models. Meanwhile, conventional methods with models or assumptions that do not fit the observations may induce deficiencies. Thereby, the development of effective model-based decompositions has received considerable attention and many advances have been reported. This article aims to provide a review for these notable advances, mainly including the incorporation of orientation compensation processing, nonnegative eigenvalue constraint, generalized scattering models, complete information utilization, full-parameter inversion schemes, and fusion of polarimetry and interferometry. Airborne Pi-SAR data sets are used for demonstration. Besides, natural disaster damage evaluation using model-based decomposition is carried out based on advanced land-observing satellite/phased array type L-band synthetic aperture radar (ALOS/PALSAR) data. Finally, further development perspectives are presented and discussed.

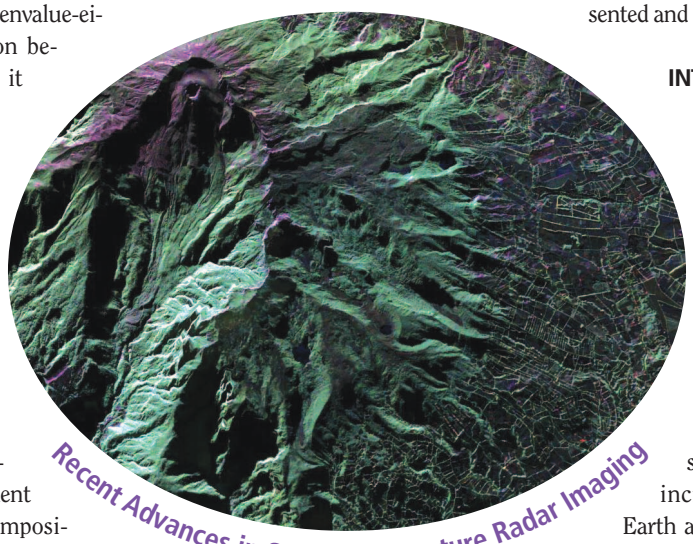


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Recent Advances in Synthetic Aperture Radar Imaging

INTRODUCTION

Microwave remote sensing has been rapidly developing over the past few decades with progresses in hardware systems, signal processing techniques, and interpretation theories. Microwave remote sensors, which can work day and night and are nearly unaffected by weather and atmospheric conditions, play an increasingly important role for Earth and other planet monitoring in both global and regional scales. PolSAR is one of the mainstream imaging systems in microwave remote sensing. It can obtain fully polarimetric information by transmitting and receiving microwaves with specific polarization states [1]–[3]. Fully polarimetric information is sensitive to scattering mechanisms related to target responses during a backscattering procedure. Full polarization acquisition can enhance the radar capability in scattering mechanism understanding and target parameters (e.g., material, shape, and orientation) retrieval. In this vein, many current airborne and spaceborne synthetic aperture

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radar (SAR) systems are equipped with a full-polarization mode. A number of PolSAR data sets have been acquired by these systems especially the routinely operated spaceborne systems such as on-orbit Radarsat-2 and TerraSAR, the retired ALOS/PALSAR, etc. [1]. These valuable data sets greatly stimulate the studies in both theory and application. Scattering mechanism understanding is a bridge between the collected data and real applications. Generally, scattering mechanisms are determined by a set of factors such as radar frequency, radar bandwidth, illumination direction, and target dielectric/geometric properties. Although the basic theory of radar polarimetry has been well established [1]–[3], scattering mechanism modeling and interpretation are still ongoing. How to effectively extract robust and useful information from collected data is still challenging.

PolSAR IS ONE OF THE MAINSTREAM IMAGING SYSTEMS IN MICROWAVE REMOTE SENSING.

In past decades, many scientists and researchers have been involved in this field to interpret the scattering mechanisms present in PolSAR data [1]–[4]. Polarimetric target decomposition theorems [5], originated from Huynen's work in the 1970s [6], became one of the most powerful and the most popularly used techniques. With this technique, a measured polarimetric matrix can be decomposed into a summation of several canonical scattering mechanisms. Thereby, the dominant scattering mechanism can be determined and physical parameters can be retrieved. A number of successful applications, such as land-cover classification and target detection [1], [4], soil moisture estimation [2], [7], forest study [1], [3], ocean study [1], agriculture study [1], glacier study [8], and natural disaster study [9]. Since the measured scattering matrix is commonly affected by speckle phenomenon [1], incoherent analysis using the second-order statistics such as the coherency/covariance matrix is mostly used. Incoherent decompositions can be divided into two categories: eigenvalue-eigenvector-based decomposition [1], [3], [5] and model-based decomposition [10], [11]. Eigenvalue-eigenvector-based decomposition becomes relatively mature since it has clearer mathematical background and has only one decomposition solution. Polarimetric entropy, mean alpha angle $\bar{\alpha}$, and polarimetric anisotropy are the most popular parameters derived from this technique and are widely used for scattering mechanism understanding [1], [3], [5]. In contrast, with various combination of scattering models, model-based decompositions can obtain different decomposition solutions. Meanwhile, conventional model-based decompositions with models or assumptions that do not fit the observations may induce deficiencies, such as the appearances of negative powers for scattering mechanisms, overestimations of volume scattering, scattering mechanism ambiguities, etc. Thereby, the development of effective model-based decompositions has received considerable attention [12]. The Freeman–Durden decomposition [10] is the pioneer of incoherent model-based decompositions. Thereafter, to enhance the scattering mechanism interpretation performance, many notable advances have been achieved, including orientation compensation strategy [13]–[15]; nonnegative eigenvalue constraint [16]–[18]; generalized scattering models [7], [19]–[26]; complete information utilization [18], [23], [26], [27]; full-parameter inversion technique [20], [21], [26]; and polarimetric-interferometric decomposition [21], [24], [25].

In 1996, Cloude and Pottier published an excellent review for target decomposition theorems [5]. Reference [12] also provides a short overview. Based on these papers, this article will investigate all these recent advances, present a brief summary, and discuss the perspectives for future developments.

BASICS OF SCATTERING MECHANISM MODELING

REPRESENTATION OF PolSAR DATA

For PolSAR, the acquired full polarimetric information can be expressed in the form of the scattering matrix, as

$$S = \begin{bmatrix} S_{\mathrm{HH}} & S_{\mathrm{HV}} \\ S_{\mathrm{VH}} & S_{\mathrm{VV}} \end{bmatrix}, \tag{1}$$

where S_{HV} is the backscattered return from vertical transmitting and horizontal receiving polarization.

Subject to the reciprocity condition ($S_{\mathrm{HV}} = S_{\mathrm{VH}}$), the Pauli scattering vector is $k_P = (1/\sqrt{2})[S_{\mathrm{HH}} + S_{\mathrm{VV}} \;\; S_{\mathrm{HH}} - S_{\mathrm{VV}} \;\; 2S_{\mathrm{HV}}]^T$. When rotated with angle θ around the radar line of sight (LOS), it becomes [1]

$$k_P(\theta) = R_3(\theta) k_P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 2\theta & \sin 2\theta \\ 0 & -\sin 2\theta & \cos 2\theta \end{bmatrix} k_P. \tag{2}$$

The coherency matrix T without rotation is

$$T = \langle k_P k_P^H \rangle = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix}, \tag{3}$$

where $\langle \;\; \rangle$ denotes the sample average, and k_P^H is the conjugate transpose of k_P. T_{ij} is the (i, j) entry of T.

The coherency matrix T with rotation angle θ is [1]

$$T(\theta) = k_P(\theta)\, k_P^H(\theta) = R_3(\theta)\, T R_3^H(\theta). \tag{4}$$

BASIC SCATTERING MODELS

The general principle of incoherent model-based decomposition is to decompose a polarimetric matrix into a summation of a set of basic scattering models. Using the coherency matrix as an example, basic scattering models include volume scattering T_{vol}, double-bounce scattering T_{dbl}, odd-bounce scattering T_{odd}, and helix scattering T_{hel}. In practice, volume scattering can represent the scattering mechanism from areas with a number of elemental scatterers inside, such as vegetation canopy and ice. Double-bounce scattering indicates dihedral

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corner reflection formed by ground-wall structures and ground-trunk interactions. Odd-bounce scattering can describe both single-bounce scattering from rough surfaces such as grass area and ocean, and triple-bounce scattering from ground-wall-ground and ground-trunk-ground structures. Helix scattering fits the reflection asymmetry condition valid in heterogeneous areas, such as complicated shape targets and built-up areas. Decomposition framework can be presented as

$$T = f_v T_{\text{vol}} + f_d T_{\text{dbl}} + f_s T_{\text{odd}} + f_c T_{\text{hel}} + \cdots, \tag{5}$$

where f_v, f_d, f_s, and f_c are model coefficients.

The number of model components is tunable for certain decomposition. With the determination of the power of each component, scattering mechanisms can be understood and applications can be explored thereafter.

VOLUME SCATTERING MODEL

Volume scattering is usually modeled by a cloud of elemental scatterers with some orientation defined by an angle θ. The coherency matrix for volume scattering is obtained by the integration with a probability density function (PDF) $p(\theta)$, and can be generally expressed as

$$T_{\text{vol}} = \int_0^{2\pi} T(\theta) p(\theta) d\theta = \begin{bmatrix} a & d & e \\ d^* & b & f \\ e^* & f^* & c \end{bmatrix}. \tag{6}$$

For randomly oriented thin dipoles, if the PDF is assumed to be uniform $(p(\theta) = 1/2\pi)$ [10], the corresponding model is

$$T_{\text{vol1}} = \frac{1}{4}\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{7}$$

With another PDF $(p(\theta) = 1/2\sin\theta)$ proposed in [11], two models can be obtained using horizontal and vertical dipoles

$$T_{\text{vol2}} = \frac{1}{30}\begin{bmatrix} 15 & 5 & 0 \\ 5 & 7 & 0 \\ 0 & 0 & 8 \end{bmatrix} \text{ and } T_{\text{vol3}} = \frac{1}{30}\begin{bmatrix} 15 & -5 & 0 \\ -5 & 7 & 0 \\ 0 & 0 & 8 \end{bmatrix}. \tag{8}$$

DOUBLE-BOUNCE SCATTERING MODEL

The double-bounce scattering is basically modeled by scattering from a dihedral corner reflector, such as ground-wall structures and ground-trunk interactions, where the reflector surfaces can be made of different dielectric materials. The corresponding coherency matrix [1] is

$$T_{\text{dbl}} = \begin{bmatrix} |\alpha|^2 & \alpha & 0 \\ \alpha^* & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \tag{9}$$

where α is complex valued and $|\alpha| < 1$ [1].

ODD-BOUNCE SCATTERING MODEL

The odd-bounce scattering is represented by a Bragg surface scattering model for slightly rough surfaces for which the depolarization effect and the cross-polarization component is negligible. The corresponding coherency matrix [1] is

$$T_{\text{odd}} = \begin{bmatrix} 1 & \beta^* & 0 \\ \beta & |\beta|^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \tag{10}$$

where $|\beta| < 1$ and β is real valued from most natural surfaces [1].

HELIX SCATTERING MODEL

The roll-invariant helix scattering component was proposed in [11] to fit the reflection asymmetry condition, such as in built-up areas. The corresponding coherency matrix is

$$T_{\text{hel}} = \frac{1}{2}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & \pm j \\ 0 & \mp j & 1 \end{bmatrix}. \tag{11}$$

CONVENTIONAL THREE- AND FOUR-COMPONENT DECOMPOSITIONS

The Freeman–Durden decomposition [10] is the first three-component incoherent model-based decomposition with the formulation as

$$T = f_v T_{\text{vol}} + f_d T_{\text{dbl}} + f_s T_{\text{odd}}. \tag{12}$$

The Freeman–Durden decomposition assumes the reflection symmetry condition for which the cross-correlation between copolarization and cross-polarization are always zero ($T_{13} = 0$ and $T_{23} = 0$). In this way, there are five known observables (T_{11}, T_{22}, T_{33}, and complex valued T_{12}) with six unknown model parameters f_v, f_d, f_s, β, and complex-valued α) to be determined. From the adopted models (7), (9), and (10), only the volume scattering contributes to the cross-polarization term T_{33}. Thereby, the volume model coefficient f_v can be determined at first and its contribution is subtracted

$$T_{\text{remainder}} = T - f_v T_{\text{vol}} = f_d T_{\text{dbl}} + f_s T_{\text{odd}}. \tag{13}$$

Based on the sign of $\text{Re}(\langle S_{\text{HH}} S_{\text{VV}}^* \rangle)$, the dominance of double- or odd-bounce scattering in the remainder matrix $T_{\text{remainder}}$ can be determined. If $\text{Re}(\langle S_{\text{HH}} S_{\text{VV}}^* \rangle)$ is positive, parameter α of model (9) is fixed as –1, otherwise parameter β of model (10) is fixed as 1. In this vein, all the model parameters can be determined.

The Yamaguchi decomposition [11] includes helix scattering model to remove the reflection symmetry assumption since it is not always valid for various land covers

$$T = f_v T_{\text{vol}} + f_d T_{\text{dbl}} + f_s T_{\text{odd}} + f_c T_{\text{hel}}. \tag{14}$$

Another improvement is the extension of volume scattering modeling. For instance, a PDF for elemental scatterer orientation

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and another two volume scattering models are introduced, shown in (8). The most suitable volume model is determined by the ratio of copolarization terms. Since only helix scattering contribute to $\mathrm{Im}[T_{23}]$, its model coefficient f_c is determined and its contribution is subtracted with the highest priority. The inversion strategy then follows the Freeman–Durden decomposition procedure.

THE GENERAL PRINCIPLE OF INCOHERENT MODEL-BASED DECOMPOSITION IS TO DECOMPOSE A POLARIMETRIC MATRIX INTO A SUMMATION OF A SET OF BASIC SCATTERING MODELS.

EXISTING PROBLEMS

Freeman–Durden and Yamaguchi decompositions are the pioneer model-based decomposition techniques. They are popularly used due to the easy physical interpretation of the scene they provide, together with their simplicity, computational efficiency, and relatively good performance. However, some issues exist too. The first is the occurrence of negative powers for double- or odd-bounce scattering after the subtraction of volume or helix scattering components. Although scattering powers can be forced to be zero or positive, these decomposition results are physically incorrect. Intrinsically, it means the scattering models do not fit the observations. The second is that double- and odd-bounce scattering components are modeled with zero cross-polarization terms. In a real situation, the depolarization effect and polarization state rotation from terrain slopes and oriented buildings can induce significant cross-polarization power. Therefore, especially oriented man-made structures and hills are usually misjudged as volume scattering dominant, which produces serious scattering mechanism ambiguity. The volume scattering models T_{vol1}, T_{vol2}, and T_{vol3} from (7)–(8) all have constant model parameters. Therefore, using (6), the volume scattering contribution is fully determined by these constants and the cross-polarization term

$$P_v = (a+b+c)f_v = \left(1+\frac{a+b}{c}\right)T_{33} = F_v T_{33}. \tag{15}$$

The dynamic range is limited and the volume scattering contribution is easily overestimated. Moreover, for parameter inversion, the implicit assumption of model priority is included, since the observations are used to satisfy helix and volume components first. In addition, conventional methods do not use the full information of a coherency matrix for modeling. For example, all models (7), (9), and (10) used in the Freeman–Durden decomposition [10] do not account for elements T_{13} and T_{23}, while elements T_{13} and $\mathrm{Re}[T_{23}]$ are not utilized in the Yamaguchi decomposition [11].

RECENT ADVANCES IN MODEL-BASED DECOMPOSITION

This section reviews recent advances in model-based decomposition and the key features are summarized in Table 1.

ORIENTATION COMPENSATION PROCESSING

Polarimetric responses of the same target with different orientations can be quite different due to the rotation of polarization state. This phenomenon causes ambiguity for scattering mechanism discrimination. Orientation compensation [1], [14], also named *deorientation processing* [6], can mitigate this orientation dependence effect by polarimetric matrix rotation around radar LOS until the cross-polarization term is minimized.

Orientation compensation was incorporated into model-based decomposition [13]–[15]. The general orientation compensation effect is to minimize the cross-polarization term T_{33}. From (15), the decomposed volume scattering power is consistently reduced. Therefore, orientation compensation can partially cure the overestimation of volume scattering power and the occurrence of negative powers. Besides, it is possible to compensate the target orientation effect on the perpendicular plane to the radar LOS using this technique. However, since there is an intersection angle between the rotation plane of oriented buildings or terrain slopes and the perpendicular plane, complete orientation compensation cannot be achieved only using this technique, which deserves further efforts.

NONNEGATIVE EIGENVALUE CONSTRAINT

The appearance of negative power due to the subtraction of the scattering component is an important issue of conventional methods. Van Zyl et al. [16] recognized this fact and proposed the nonnegative eigenvalue decomposition to theoretically avoid the occurrence of nonphysical negative power. Using three-component decomposition as an example, they introduced the nonnegative eigenvalue constraint (NNEC) during the subtraction of the volume scattering component. The core idea is to guarantee the eigenvalues of remainder matrix $T_{\mathrm{remainder}}$ are nonnegative. Then, the subtraction processing will not produce negative power. Using NNEC, the maximum volume model coefficient can be determined

$$T_{\mathrm{remainder}} = T - f_v^{\max} T_{\mathrm{vol}}. \tag{16}$$

Therefore, when $f_v \le f_v^{\max}$ is satisfied, the negative power can be avoided. With the reflection symmetry condition, the analytical expression of $f_v^{\max}$ is derived in [16]. More recently, its analytical expression is extended to the nonreflection symmetry condition using the principle minor theory [17]. The optimal determination of f_v is still open. In [3], a generalized Freeman–Durden decomposition is established and the appearance of negative powers is theoretically avoided using both the eigenvalue technique and the orthogonality of double- and odd-bounce scattering models. Furthermore, with certain conditions, the idea of NNEC is generalized and volume model coefficient f_v can be uniquely determined [18].

OTHER VOLUME SCATTERING MODELS

To better fit the elemental scatterer orientation and to extend the dynamic range of volume scattering model, other PDFs have been utilized [21]–[23]. The von Mises distribution has been used in [21] to characterize a vegetation canopy. The nth

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[TABLE 1] A SUMMARY OF TYPICAL MODEL-BASED DECOMPOSITIONS.

METHODS	INFORMATION UTILIZATION	NEGATIVE POWER	MODEL INVERSION PRIORITY	COMPUTATION EFFICIENCY	NOTABLE REMARKS
FREEMAN–DURDEN DECOMP. [10]	T_{13}, T_{23} UNACCOUNTED	YES	VOLUME>DOUBLE-/ ODD-BOUNCE	HIGH	BASIC SCATTERING MODELS
FREEMAN II DECOMP. [19]	T_{13}, T_{23} UNACCOUNTED	YES	VOLUME>DOUBLE/ ODD-BOUNCE	HIGH	VOLUME SCATTERING MODEL IS FITTED BY A SHAPE PARAMETER
YAMAGUCHI DECOMP. [11]	T_{13}, $RE[T_{23}]$ UNACCOUNTED	YES	HELIX>VOLUME> DOUBLE-/ ODD-BOUNCE	HIGH	INTRODUCE THE FOURTH HELIX COMPONENT AND TWO MORE VOLUME SCATTERING MODELS
YAMAGUCHI (FREEMAN-DURDEN) DECOMP. + ORIENTATION COMPENSATION [13]–[15]	T_{13} (T_{13}, $IM[T_{23}]$) UNACCOUNTED	YES	HELIX>VOLUME> DOUBLE-/ ODD-BOUNCE	HIGH	INCORPORATE THE ORIENTATION COMPENSATION TO MINIMIZE T_{33} AND ELIMINATE $RE[T_{23}]$
GENERALIZED FREEMAN-DURDEN DECOMP. [3]	T_{13}, T_{23} UNACCOUNTED	NO	VOLUME>DOUBLE-/ ODD-BOUNCE	HIGH	HYBRID FREEMAN–DURDEN/ EIGENVALUE TECHNIQUE
VAN ZYL ET AL. DECOMP. [16]	T_{13}, $IM[T_{23}]$ UNACCOUNTED	NO	VOLUME>DOUBLE-/ ODD-BOUNCE	HIGH	INCORPORATE THE NNEC TO THEORETICALLY AVOID THE NEGATIVE POWER
GULAB ET AL. DECOMP. [27]	FULL POLARIMETRIC INFORMATION	YES	HELIX>VOLUME> DOUBLE-/ ODD-BOUNCE	HIGH	DOUBLE UNITARY ROTATION PROCESSING
CUI ET AL. DECOMP. [18]	FULL POLARIMETRIC INFORMATION	NO	VOLUME>DOUBLE-/ ODD-BOUNCE	HIGH	HYBRID DECOMPOSITION USING BOTH SCATTERING MODELS AND EIGENDECOMPOSITION
LEE ET AL. DECOMP [23]	T_{13}, $IM[T_{23}]$ UNACCOUNTED	VERY LIMITED	VOLUME>DOUBLE-/ ODD-BOUNCE	HIGH	GENERALIZED DOUBLE- AND ODD-BOUNCE OR VOLUME SCATTERING MODELS
ARRI ET AL. DECOMP [20], [22]	T_{13} UNACCOUNTED OR FULL POLARIMETRIC INFORMATION	NO	PARTIAL PRIORITY, SIMULTANEOUSLY INVERSION	LOW	GENERALIZED VOLUME SCATTERING MODELS, FULL-PARAMETER INVERSION
CHEN ET AL. DECOMP. [26]	FULL POLARIMETRIC INFORMATION	NO	NO PRIORITY, SIMULTANEOUSLY INVERSION	LOW	GENERALIZED DOUBLE- AND ODD-BOUNCE MODELS, FULL-PARAMETER INVERSION
NEUMANN ET AL. DECOMP. [21]	POLARIMETRIC AND INTERFEROMETRIC INFORMATION	NO	NO PRIORITY, SIMULTANEOUSLY INVERSION	LOW	GENERALIZED MODELS ESPECIALLY FOR VEGETATIONS, FULL-PARAMETER INVERSION
DAVID BALLESTER–BERMAN ET AL. DECOMP. [24]	POLARIMETRIC AND INTERFEROMETRIC INFORMATION	NO	VOLUME>DOUBLE-/ ODD-BOUNCE	LOW	OBTAIN THE POWER AND LOCATION OF EACH SCATTERING MECHANISM AT THE VERTICAL DIMENSION OF THE SCENE
POLARIMETRIC-INTERFEROMETRIC DECOMP. [25]	POLARIMETRIC AND INTERFEROMETRIC INFORMATION	NO	VOLUME>DOUBLE/ ODD-BOUNCE	HIGH	USING POLARIMETRIC AND INTERFEROMETRIC INFORMATION TO FIT VOLUME SCATTERING

power cosine-squared distribution, including two parameters of the mean orientation angle and the randomness indicator, was developed in [22]. Meanwhile, generalized volume scattering models generated from arbitrary-shaped elemental scatterers instead of the thin dipole assumption were also proposed [22]. Besides, a family of models based on the multilook phase difference distributions, which are better suited to characterize orientation angle distributions, was proposed [23].

In addition, empirical volume scattering models have also been proposed as another alternative. An empirical model with shape parameter ρ was introduced to particularly fit randomly oriented prolate spheroidal and elliptical scatterers in [19] (notated as the Freeman II decomposition). Since polarimetric SAR interferometry (PolInSAR) coherence has a close relationship to forest structures [3], [28], volume scattering can be modeled from it and an adaptive volume scattering model was proposed [25] using PolInSAR coherence to be functioned as the shape parameter. Identity matrix with the highest entropy was also proposed to fit volume scattering [13].

GENERALIZED DOUBLE- AND ODD-BOUNCE SCATTERING MODELS

In conventional methodologies, the double- and odd-bounce scattering mechanisms are assumed not to contribute to the cross-polarization term. However, in real situations, terrain slopes in the along-track direction and/or oriented buildings rotate the polarization basis of the scattering matrix, which will induce significant cross-polarization power [1], [2], [16]. Since the conventional models do not fit these situations, the scattering mechanism ambiguity is serious. Recently, several papers addressed this issue for more generalized double- and odd-bounce scattering modeling [7], [23], [26]. In [7], the Bragg surface model has been extended as the advanced X-Bragg model to fit the cross-polarization term induced by the depolarization effect. In [23], incoherent double- or

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odd-bounce scattering models have been incorporated. Recently, in [26], double- and odd-bounce scattering models are generalized to fit the cross-polarization and off-diagonal terms, by separating their independent orientation angles θ_{dbl} and θ_{odd}.

COMPLETE INFORMATION UTILIZATION

A coherency matrix contains nine real values (three real-valued quantities and three complex-valued quantities). Each of them has specific physical meaning. If scattering mechanisms are well modeled, each element of a coherency matrix should be fitted. The three-component Freeman–Durden decomposition [10] accounts for five of these parameters. The Yamaguchi decomposition [11] introduced the fourth component and accounts for six parameters. With the orientation compensation [14], $\text{Re}[T_{23}] = 0$ is achieved and nine elements are reduced to eight. Thereby, there remains three and two unaccounted parameters in Freeman–Durden and Yamaguchi decompositions, respectively. The details of many current model-based decompositions are summarized in Table 1.

How to model the scattering mechanisms using complete information is a major motivation, and three representative approaches were presented in [18], [26], and [27]. A further modified Yamaguchi decomposition was reported [27] and accounts for all parameters based on special double unitary transformations. In [26], a general decomposition framework was proposed to utilize all elements of a coherency matrix. All model parameters are optimally obtained using the full-parameter inversion technique. A hybrid decomposition using full information was established [18]. It generalized the van Zyl et al. decomposition [16] to uniquely determine volume scattering contribution from the scattering model while using the eigendecomposition to determine the double- and odd-bounce scattering contributions.

FULL-PARAMETER INVERSION STRATEGY

Model parameters inversion is carried out at a certain order under some specific conditions for majority of the established model-based decompositions. For conventional Freeman–Durden and Yamaguchi decompositions, the volume scattering and helix scattering solely contributes to cross-polarization term T_{33} and $\text{Im}[T_{23}]$, respectively. Therefore, their contributions are subtracted from the total measured data at first. Double- and odd-bounce scattering contributions are calculated in turn from the remainder matrix thereafter with a branch condition. This strategy is the main weakness and may overestimate volume scattering contribution, induce negative power, etc. In addition, it also implies that the volume and helix scattering components share the higher priority. Especially in mixed scenes, such a priority assumption seems unreasonable.

Generalized decomposition frameworks and full-parameter inversion techniques have been proposed [20], [26] to obtain all the unknown parameters simultaneously using linear or nonlinear optimization algorithms. Basically, the decomposition framework is generalized as

$$T = f_v T_{\text{vol}} + f_d T_{\text{dbl}} + f_s T_{\text{odd}} + f_c T_{\text{hel}} + \ldots + f_r T_{\text{residual}}. \tag{17}$$

Suitable scattering models that guarantee parameter inversion as a determined problem can be included into (17). T_{residual} is regarded as the residual induced by speckle, noise, and unmodeled contributions to the data. The residual matrix T_{residual} can measure how well those models fit the observations. Therefore, for model inversion, the optimization criterion is to minimize the residual. The optimization adopted in [20] and [26] is

$$\min : \| T_{\text{residual}} \|_2, \tag{18}$$

where $\| T_{\text{residual}} \|_2$ is the L2-norm of T_{residual}.

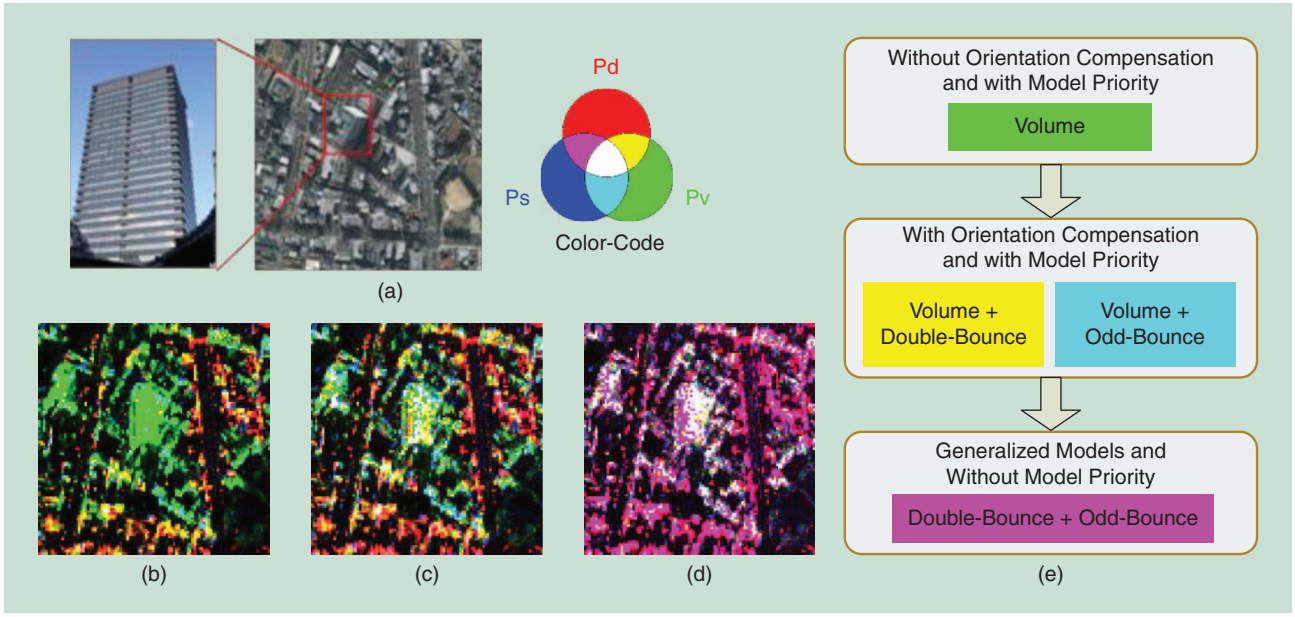
Minimization (18) can be solved by a nonlinear least squares optimization, as adopted in [26]. Full-parameter inversion can obtain all model parameters optimally and simultaneously while avoiding the model priority assumption and negative power issue. Note that nonlinear optimization takes more computation time and may obtain only local minimal of $\| T_{\text{residual}} \|_2$, which needs further studies.

THE COMPLEMENTARY INFORMATION BETWEEN POLARIMETRY AND INTERFEROMETRY HAS BEEN DEMONSTRATED TO BE VERY PROMISING FOR MANY APPLICATIONS, SUCH AS FOREST CHARACTERIZATION AND LAND-COVER CLASSIFICATION.

POLARIMETRIC-INTERFEROMETRIC DECOMPOSITION

PolInSAR, which is a combination of PolSAR and interferometric SAR, has been well established [3], [28]. This combination allows understanding different scattering mechanisms located at different heights. The complementary information (e.g., PolInSAR coherence) between polarimetry and interferometry has been demonstrated to be very promising for many applications, such as forest characterization and land-cover classification [1], [3]. PolInSAR studies contain two main aspects. The first is the development of innovative imaging techniques especially for the vertical profile or the three-dimension reconstruction of targets using single- or multibaseline PolInSAR data. The second is the physical parameters (e.g., forest height, biomass) inversion, which also covers a wide scope. Each of these important topics and recent advances deserves separate papers to discuss in depth. In the second aspect, there is a small branch utilizing the concept of model-based decomposition and the framework of (5). Only this technique branch, mainly including three reported methods [21], [24], [25], will be introduced in this section. The main purpose of these methods is to incorporate interferometric information in scattering mechanism modeling. With extra information to conventional polarimetric decomposition framework, decomposition performance can be enhanced. In [21], scattering modeling especially in vegetation areas has been intensively studied and a set of volume scattering models using single- or multibaseline PolInSAR data has been established.

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[FIG1] A demonstration with Pi-SAR PolSAR data. (a) A Google Earth optical image and a photo for the tall building, (b) and (c) Freeman–Durden decomposition without and with orientation compensation [10], [14], (d) Chen et al. decomposition [26] without orientation compensation. The decomposition images are colored by P_d (red), P_v (green), and P_s (blue). (e) Decomposition performance comparison over oriented buildings.

Detailed geophysical parameters describing the vegetation layers can be derived. In [24], the concept of the Freeman–Durden decomposition has been extended to PolInSAR polarimetric-interferometric correlation matrix by adding the interferometric responses of each scattering mechanism. Thereby, both the power and location of each scattering mechanism can be determined. In [25], an adaptive polarimetric-interferometric model-based decomposition has been developed using PolInSAR coherence. Demonstrations with real PolInSAR data validated its performance especially for oriented building characterization. These are the first attempts to fuse the polarimetric and interferometric information for scattering mechanism modeling. More efforts are needed for further study.

EXPERIMENTAL DEMONSTRATION AND INVESTIGATION

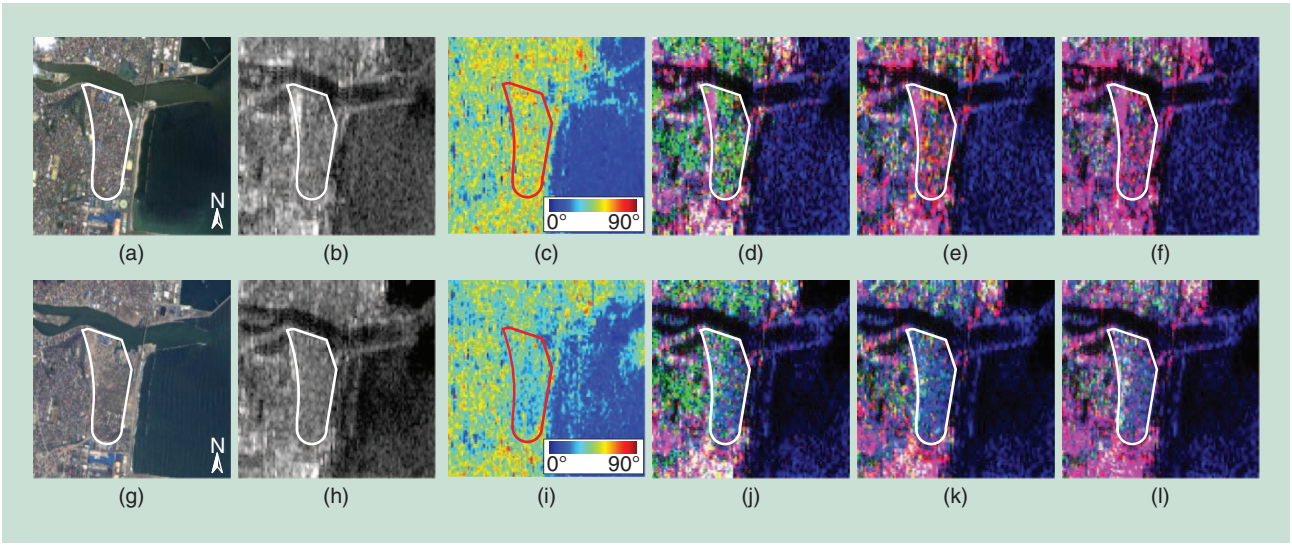
DEMONSTRATION WITH AIRBORNE Pi-SAR DATA

This example demonstrates the effects of model priority issue, orientation compensation, generalized models, and full parameter inversion scheme. Airborne Pi-SAR X-band PolSAR data covering an urban region from the downtown of Sendai, Japan, is used. The optical image and a photo for the tall building are shown in Figure 1(a). The representative three-component Freeman–Durden decomposition without and with orientation compensation [10], [14] and the recently developed Chen et al. decomposition [26] are used for comparison. The Chen et al. decomposition [26] uses the generalized double- and odd-bounce scattering models fitting the cross-polarization term and off-diagonal terms of a coherency matrix. To guarantee that the model inversion is a determined problem, a lookup table including representative volume scattering models with constant parameters is formed. Suitable volume scattering model and optimal model parameters are determined using the full-parameter inversion technique (18). Decomposition results are shown in Figure 1(b)–(d), respectively. For Freeman–Durden decomposition, volume scattering contribution is first determined and subtracted. This processing implies the model priority assumption and volume scattering contribution always exist for any target. In Figure 1, without orientation compensation, the oriented tall building is obviously judged as dominant volume scattering, which is usually exhibited by vegetated areas. Thereby, scattering mechanism ambiguity occurs. With orientation compensation, the cross-polarization term is minimized and the volume scattering contribution is reduced. As can be seen in Figure 1(c), volume scattering contribution is greatly decreased. In reality, oriented buildings also induce significant cross-polarization term. Since conventional double- and odd-bounce models (9) and (10) do not account for it, and due to the model priority issue, the volume scattering contributions over these man-made targets are still overestimated. By that

[TABLE 2] DOMINANT SCATTERING POWER CONTRIBUTION (%) FOR THE SELECTED BUILDING.

	P_d	P_v	P_s
FREEMAN–DURDEN DECOMPOSITION [10]	2	94	4
FREEMAN–DURDEN DECOMPOSITION WITH ORIENTATION COMPENSATION [14]	30	45	25
CHEN ET AL. DECOMPOSITION [26]	46	13	41

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[FIG2] (a)–(f) Before- and (g)–(l) after-tsunami ALOS data and decomposition results. (a) and (g) show the ALOS optical image, (b) and (h) illustrate HH polarization amplitude data, (c) and (i) are the mean alpha angle parameter derived from Cloude–Pottier decomposition [1], [5], (d) and (j) show the Yamaguchi decomposition without orientation compensation [11], (e) and (k) show the Yamaguchi decomposition with orientation compensation [15], and (f)–(l) shows the Chen et al. decomposition [26] without orientation compensation. The decomposition images are colored by P_d (red), P_v (green), and P_s (blue).

means, comparable volume scattering and double- or odd-bounce scattering contributions coexist for these oriented buildings, which appear yellow or cyan in the RGB composite image in Figure 1(c). In comparison, the generalized models fit the cross-polarization terms while the full-parameter inversion technique avoids the negative power and model priority assumption. The general decomposition can successfully identify these buildings as man-made structures. In Figure 1(d), these buildings show purple since both double- and odd-bounce scattering mechanisms are dominant. For the tall building shown in Figure 1(a), quantitative examination is also carried out and summarized in Table 2. Compared to the original Freeman–Durden decomposition [10], with orientation compensation, dominant volume scattering obviously reduces from 94% to 45%, while the percentages of the dominant double- and odd-bounce scattering increase from 2% to 30% and 4% to 25%, respectively. Even still, volume scattering contribution is still overestimated and this building will be misjudged as dominant volume scattering. In comparison, dominant volume scattering further reduces to 13%, while dominant double- and odd-bounce scattering increase up to 46% and 41%. This building can be successfully discriminated as a man-made structure, which validates the efficiency of the advances.

APPLICATION FOR NATURAL DISASTER EVALUATION

The occurrence of observed natural disasters, such as earthquakes and tsunamis, appears to have increased in recent decades [9]. With the advantages of a large imaging scene, revisiting capability, and multitemporal data archives, spaceborne SAR sensors play an extremely important role for natural disaster evaluation. The following demonstration uses the spaceborne ALOS data over the seriously damaged city of Ishinomaki, induced by the Eastern Japan great earthquake and tsunami that happened 11 March 2011. The after-tsunami optical and PolSAR data sets were acquired on 8 April 2011, while the latest before-tsunami data sets were acquired on 21 November 2010. Optical images are shown in Figure 2(a) and (g). The Pacific Ocean is located to the right of the images. Buildings in the region, highlighted by the line, were almost completely flushed away leaving a relatively rough surface. Besides, the orientations of these buildings are inclined to the sensor flight pass.

SPACEBORNE SAR SENSORS PLAY AN EXTREMELY IMPORTANT ROLE FOR NATURAL DISASTER EVALUATION.

From scattering interpretation theory, this built-up area should response dominant double-bounce scattering induced by the ground-wall structures before the damage. However, after the tsunami, most of these buildings were swept away and the ground-wall structures were dramatically decreased. Therefore, this seriously damaged area should be obviously changed into dominant odd-bounce (surface) scattering. As a comparison, the amplitudes of HH polarization are shown in Figure 2(b) and (h). Visually, single polarization is less sensitive to the changes of target structures, compared with fully polarimetric features. Meanwhile, mean alpha angle $\bar{\alpha}$, which is the primary parameter for scattering mechanism indication in the Cloude–Pottier decomposition [1], [5], is derived and shown in Figure 2(c) and (i). For before-tsunami cases, $\bar{\alpha} \rightarrow 90°$ and indicates double-bounce scattering, while $\bar{\alpha} \rightarrow 0°$ and indicates surface scattering for after-tsunami cases. Therefore, the $\bar{\alpha}$

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parameter can potentially discriminate damaged buildings. The Yamaguchi decomposition without and with orientation compensation [11], [15] and the Chen et al. decomposition [26] are also used for demonstration, shown in Figure 2(d)–(f) for before-tsunami cases and Figure 2(j)–(l) for after-tsunami cases, respectively. After the tsunami, the damaged area is correctly judged as dominant odd-bounce scattering (blue) indicating a rough surface from all three methods. However, for before-tsunami cases, the overestimation of volume scattering is significant without orientation compensation and this built-up area is falsely determined as dominant volume scattering (green), shown in Figure 2(d). In comparison, the other two approaches can successfully identify them as man-made structures that exhibit dominant double-bounce scattering (red). These polarimetric features are more stable during repeated data collects. The reduction of dominant double-bounce scattering amount has been verified to have a linear relationship with the amount of damaged ground-wall structures and has been utilized for quantitative evaluation of urban damage [9].

SINCE THE SCATTERING MECHANISM CLOSELY RELATES TO LOCAL INCIDENCE ANGLE AND TARGET ORIENTATION, USING MULTILOOKINGDIRECTION DATA SETS IS ANOTHER PROMISING TECHNIQUE.

DISCUSSIONS AND PERSPECTIVES

PolSAR DATA PREPROCESSING ISSUE

There are two main aspects in PolSAR data preprocessing. The first is the data calibration. Generally, the acquired PolSAR data is affected by the channel imbalance and cross-talk effects that distort the polarization purity [2], [3]. Polarimetric calibration should be implemented at first for further processing. The aforementioned polarimetric target decompositions are all based on well-calibrated data. For low-frequency band (e.g., P-, L-band) spaceborne systems, the Faraday rotation effect induced by the ionosphere should also be compensated beforehand. Propagation distortions including Faraday rotation effect in model-based decomposition have been investigated in [3].

The second aspect is speckle reduction. Speckle phenomenon is an intrinsic property for coherent imaging system of PolSAR [1]. The existence of speckle phenomenon makes information extraction more difficult. Speckle reduction is also a preprocessing step for PolSAR data utilization. Suitable speckle filter should significantly reduce speckle phenomenon while preserving image details [1].

RADAR FREQUENCY ISSUE

Target backscattering properties usually vary with the illumination frequency due to its physical characteristics, especially the geometric features (e.g., size, shape, and structure) with respect to the radar wavelength. Meanwhile, low frequencies such as P- and L-band usually have a penetration depth into forest/vegetation canopies, deserts, dry bare soil areas, etc. Intrinsically, incident microwaves with different wavelengths interact with different parts of a complex target or medium and the polarimetric responses may differ accordingly. Thereby, scattering mechanisms of a same target strongly depend on the observation frequency and so are the interpretation results from model-based decomposition. For example, the scattering mechanism from vegetations may totally differ at the P- and X-band. In addition, the scattering models presented in the sections "Basics of Scattering Mechanism Modeling" and "Recent Advances in Model-Based Decomposition" have their theoretical assumptions that should be well satisfied before utilization.

HIGH SPATIAL RESOLUTION ISSUE

With the advances in SAR imaging systems, fine-resolution data in the order of decimeter and centimeter can be available. At such resolutions, scatterers may be very limited. Some of the current modeling schemes should be updated. For example, modeling volume scattering by a cloud of oriented elemental scatterers should be reconsidered. Besides, note that incoherent polarimetric decomposition with sample average of measured data may blur image detail, while coherent polarimetric decomposition with scattering matrix (1) have only five freedoms for modeling. Thereby, how to fit these fine-resolution data provides challenges to scattering mechanism modeling and interpretation. In another aspect, more detailed information such as opening or closing a window may be sensed with high spatial resolution systems. Suitable decomposition techniques are highly expected for these situations.

PERFORMANCE EVALUATION ISSUE

A number of advanced model-based decompositions have been reported. They raise urgent questions: how to judge whether one proposed scattering model is reasonable or superior and how to fairly evaluate the performance? Actually, there is no standard answer. From the literature and according to our studies, the used evaluation criterion are summarized and discussed:

1) *Reflection symmetry assumption*. This assumption is mainly satisfied in vegetated areas with relatively flat topography. More generalized decomposition should also account for the reflection asymmetry condition.

2) *Information utilization*. The number of the used quantities of a polarimetric matrix is a good measure to evaluate a model-based decomposition proposal. Complete information utilization will be a mainstream concept for future development.

3) *Negative power occurrence*. This is an important issue since the decomposition results are nonphysical. Intrinsically, it indicates that the scattering models do not fit the observations.

4) *Scattering mechanism discrimination*. This is one key criterion to evaluate the performance of model-based decomposition. The majority of recent advances are also motivated to

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overcome the scattering mechanism ambiguity between forests and oriented buildings.

5) *Model priority assumption*. For mixed land covers, there is no proper reason to assign the model priority. If the volume scattering component is determined at first, each scatterer will have volume scattering contribution. For oriented buildings, this contribution is not negligible. Generally, it is more reasonable to consider each scattering model share the equivalent priority.

6) *Computation efficiency*. Computation time and implementation complexity are considered in real applications. Basically, computationally effective methods are preferred.

These criteria can be the basics for model-based decomposition performance evaluation and comparison. Table 1 provides a summary of these criteria for typical model-based decompositions. Note that it is not reasonable to claim one proposed decomposition is always superior to others. In practice, how to choose the most suitable decomposition is also data oriented and application oriented.

FURTHER GENERALIZED MODELING

One difficulty in further development of model-based decomposition is how to balance model generalizations and model parameters inversion. A coherency matrix, providing nine inputs, allows at most nine unknown parameters for scattering mechanism modeling. There are several potential ways for future exploration. The first is still based on polarimetric data and segments the land cover beforehand. Specifically generalized scattering models are established for different terrains using more freedom in the modeling. For example, for pure built-up areas with dominant double-bounce scattering, ocean and grass areas with dominant odd-bounce scattering, more generalized double- and odd-bounce models can be founded accordingly. The core idea is that it may not be necessary to keep three or four scattering components for known land covers, and more detailed information can be extracted thereafter. The second method is to develop more generalized scattering models that do not reduce the scattering components by fusing additional data sets. The polarimetric-interferometric decomposition [21], [24], [25] is one attempt. Since the scattering mechanism closely relates to local incidence angle and target orientation, using multilooking-direction data sets is another promising technique. Furthermore, a combination of the aforementioned techniques may also be investigated in further studies.

CONCLUSIONS

Recent advances in scattering modeling and model-based decomposition theorem were reviewed. The notable achievements include orientation compensation processing, nonnegative eigenvalue constraint, generalized scattering models, complete information utilization, full-parameter inversion strategy, and the polarimetric-interferometric decomposition scheme. These advances contribute to make scattering models more adaptive, better fit observations and guarantee physically meaningful decomposition solutions. The key features of these advances have been summarized. Performance evaluation and further development perspectives were also discussed. One promising way is to fuse multiple data to better model scattering mechanisms, such as the polarimetric-interferometric modeling attempts. Besides, with the progress in PolSAR sensors, imaging modes (e.g., bistatic, hybrid-polarization and multi-incident-angle modes) and application requirements, the development of specific scattering mechanism interpretation techniques, multiangular decomposition [29], and compact/hybrid decomposition [30] techniques are also highly preferred.

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